

IN SPACE & TIME

With 368 private space tech firms, India's space economy is the fifth largest globally after the US, the UK, Canada and Germany

► India constitutes nearly 2% of the \$360-billion commercial space economy, which is expected to increase to over 10% by 2030

► Since June 2020, private enterprises are allowed to participate in all areas of the space sector economy

► Currently, around 75 start-ups are registered under the space technology category on the Startup India portal

► Indian National Space Promotion and Authorisation Centre (IN-SPACe), formed in June 2020, supports private players to become independent actors instead of just vendors

► Indian Space Association (ISpA) was launched in October 2021 to accelerate the development of the commercial space sector

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University, chuckling. Thus, besides sharing technology and expertise, the mechanism being evolved also has provisions for promotion and handholding to help the new entrants find their feet in the market.

Also, while there is a lot of activity in areas like launch vehicles or satellites, the number of entrants in applications is currently very limited. The development of applications based on space exploration activities is important for the sector's growth. "The big money is not in rockets or satellites, but in space applications, which manifest themselves in varied domains like agriculture, medicine, Internet of Things (IoT) and geospatial. We are expecting many more new start-ups to enter those areas," says Lt. Gen. Anil Kumar Bhatt, Director General, Indian Space Association (ISpA), a forum for private participants in the sector that was formed in 2021.

Here, a company like Tathya Earth stands out. The Navi Mumbai-based start-up was established in 2019 by IIT Madras technology graduate Gunjan Thakuria, 35, and three partners to offer predictive economic forecasts based on satellite imagery. The idea first occurred to them while working on a problem statement, when they came across research work by Stanford and Harvard scientists involving satellite images to analyse poverty levels and economic activi-

ties in data-poor regions of Sub-Saharan Africa. "The World Bank was also studying how their projects were impacting those societies. Since it was difficult to get field data, they were trying out novel approaches like satellite images to do that. That is when we realised how satellite images could be used to analyse economic activities," explains Thakuria.

The company focusses on using alternative data sets obtained through satellite imaging and technology to study economic trends. The team at Tathya started off with the niche metals and mining space, one of the core pillars of the global economy. The company applies advanced algorithms on satellite images to map the entire supply chains, for instance, to predict how much steel is being produced at multiple steel mills or copper ore extracted from mines in different parts of the world. The company promoters are also looking at utilising their algorithms and platform in areas such as oil & gas, power and emissions. "As we ourselves subscribe to data sets from multiple market sources, we understand there are often problems with them. Moreover, they are often only available long after an event is over. So, we thought why not use satellite data to understand what is happening in near real time and not just rely on a company or government declared compass," says Thakuria.

Given the sensitivity of this sector, the government is expected to maintain a close vigil on developments and, therefore, it will take a few more years before significant liberalisation of the sector could happen, say legal experts. "The key challenge that will need to be addressed going forward will be to provide adequate technical infrastructure to meet the growing requirements of the private sector, and make available domestic funding to avoid commercial exploitation by foreign players through foreign funding," says Hemang Parikh, Partner at law firm DSK Legal.

Besides, to fast-track things, the Space Activity Bill to regulate and promote private participation in the sector must become law at the earliest. "That would help provide a clear policy guideline to private industry. Sending things to space is a highly technical matter involving issues such as national security and insurance. It is very important that approvals and other things happen very fast because time is money in this domain," says Bhatt.

At the moment it appears that the chances of India becoming a prominent commercial space hub are fairly bright. The country enjoys a clear advantage in terms of possessing a vibrant, technologically advanced and cost-efficient space sector. If leveraged well, these can help propel growth in the commercial space sector way beyond the solar system. Metaphorically speaking, of course. **BT**

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